

LAB 01: Network Modes Comparison Report



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1. Objective

This lab investigates how VirtualBox's five network adapter modes NAT, NAT Network, Bridged, Host-Only, and Internal Network affect connectivity between two virtual machines (Kali Linux and Metasploitable2), along with their reachability from the host machine and access to the internet. Each mode was configured and tested individually using live connectivity checks (ip a, ping), with results captured as screenshots rather than assumed documentation. The objective is to establish a correctly understood, deliberately isolated lab network as the foundation for all future offensive and defensive security labs, and to build a practical understanding of network isolation because misunderstanding network boundaries is one of the most common sources of both misconfigured lab environments and real-world detection blind spots.

2. Lab Environment

2.1. Host Machine:

- OS: Windows 11
- Hypervisor: Oracle VirtualBox 7.2.12

2.2.Kali Linux (Attacker/Analysis Machine):

- Version: kali-linux-2026.2-virtualbox-amd64
- RAM Allocated: 2048 MB
- CPUs: 2
- Role: Primary attacker and packet analysis machine in all future labs

2.3.Metasploitable2 (Target Machine):

- Version: Metasploitable2 (Ubuntu 8.04-based, intentionally vulnerable)
- RAM Allocated: 512 MB
- CPUs: 1
- Role: Intentionally vulnerable targets never expose this machine to a real network

3. Networking Concepts Primer

Before reading the results, these concepts must be understood clearly they are referenced throughout every mode's analysis.

3.1.What ip a show and how to read it:

ip a (short for ip address) lists every network interface on the machine along with its current IP configuration. The key lines to read are:

- eth0: the name of the first Ethernet interface. This is the "door" all network traffic passes through. A second adapter would be eth1, and so on.
- inet 10.0.2.15/24: the IPv4 address assigned to that interface. The /24 is CIDR notation meaning the first 24 bits identify the network, and the remaining 8 bits identify individual hosts. This gives 256 possible addresses (.0 to .255) on that subnet, with .0 reserved as the network address and .255 as the broadcast address.
- link/ether 08:00:27:xx:xx:xx : the MAC address (hardware-level address) of that interface. This is what ARP resolves IP addresses to. Every frame on a local network is addressed to a MAC, not an IP.
- brd ff:ff:ff:ff:ff:ff : the broadcast MAC address. Any frame sent here reaches every device on the local segment simultaneously.

3.2.Subnet mask / CIDR:

Two machines must share the same subnet to communicate directly without a router.

192.168.56.101 and 192.168.56.102 share the same /24 network (192.168.56.0) so they can communicate directly. A machine at 10.0.2.15 is on an entirely different network and would require routing to reach.

3.3.What ping does:

ping sends ICMP Echo Request packets to a target and listens for ICMP Echo Reply packets back. It tells you three things: whether a path exists to the target, how long the round trip takes (latency), and whether any packets are lost. A sub-millisecond ping to yourself is normal (loopback). A 1–10ms ping to another VM on the same host is normal. A 40–100ms ping to an internet address is normal.

3.4.What "Network is unreachable" means vs a timeout:

These are two completely different failures and it is critical to distinguish them:

- "Network is unreachable" : the sending machine's own kernel looked up its routing table and found no route to that destination at all. The packet was never sent onto the wire. This happens before any network traffic occurs.
- A timeout (no reply, but ping keeps sending) : a route exists and packets are genuinely sent, but nothing answers. The destination may not exist, may be filtering ICMP, or may be down.

4. Methodology

The following procedure was applied identically to each of the five modes:

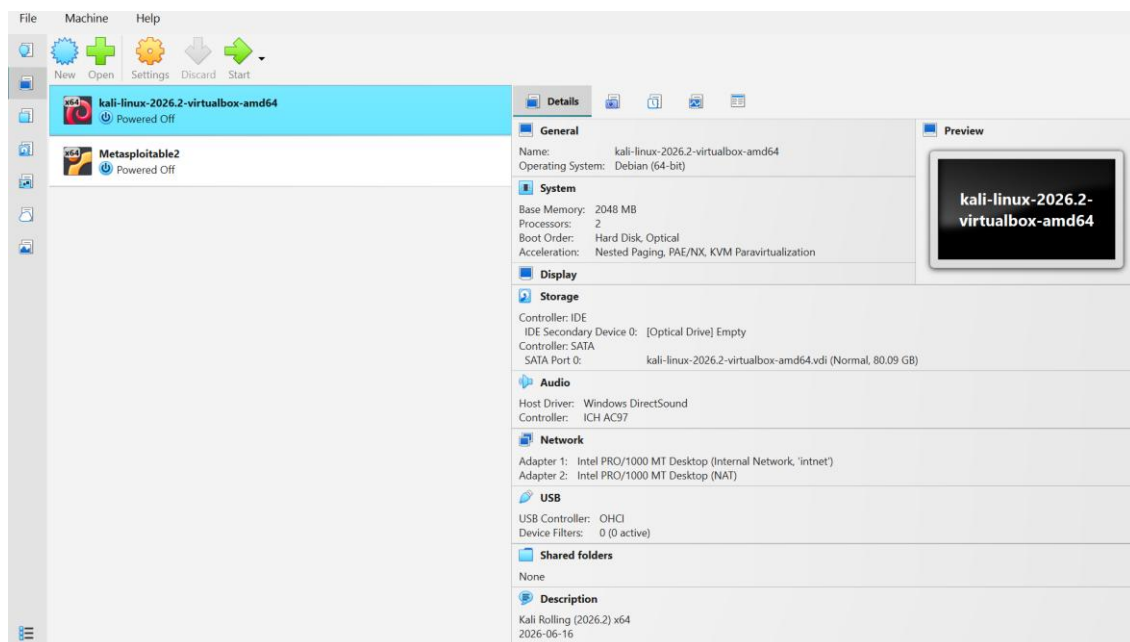
1. Power off both VMs completely (not saved state VirtualBox requires full power-off to change adapter settings).
2. Configure Adapter 1 on both Kali and Metasploitable2 to the mode under test.
3. Boot both VMs and run `ip a` on each to record the assigned IP address, subnet, and whether addressing was automatic or manual.
4. Test VM-to-VM connectivity in both directions using `ping -c 4 <target-ip>`.
5. Test internet access from each VM using `ping -c 4 8.8.8.8`.
6. Test host-to-VM reachability by pinging both VM IPs from Windows PowerShell.
7. Screenshot every step as evidence.

Special handling per mode is noted in that mode's section (e.g. NAT Network requires creating the network first; Internal Network requires manual static IP assignment since no DHCP server exists).

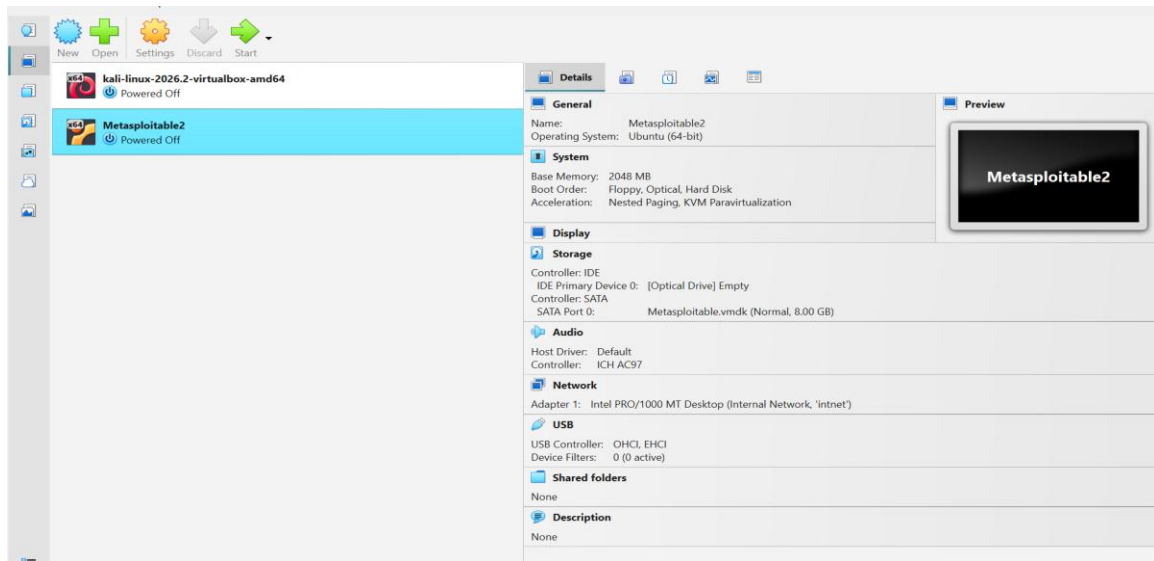
5. Setup Tasks

1. Confirm current specifications of both VMs (RAM, CPU, VirtualBox version) record these, they go in your Lab Environment section.

5.1.For kali Linux:

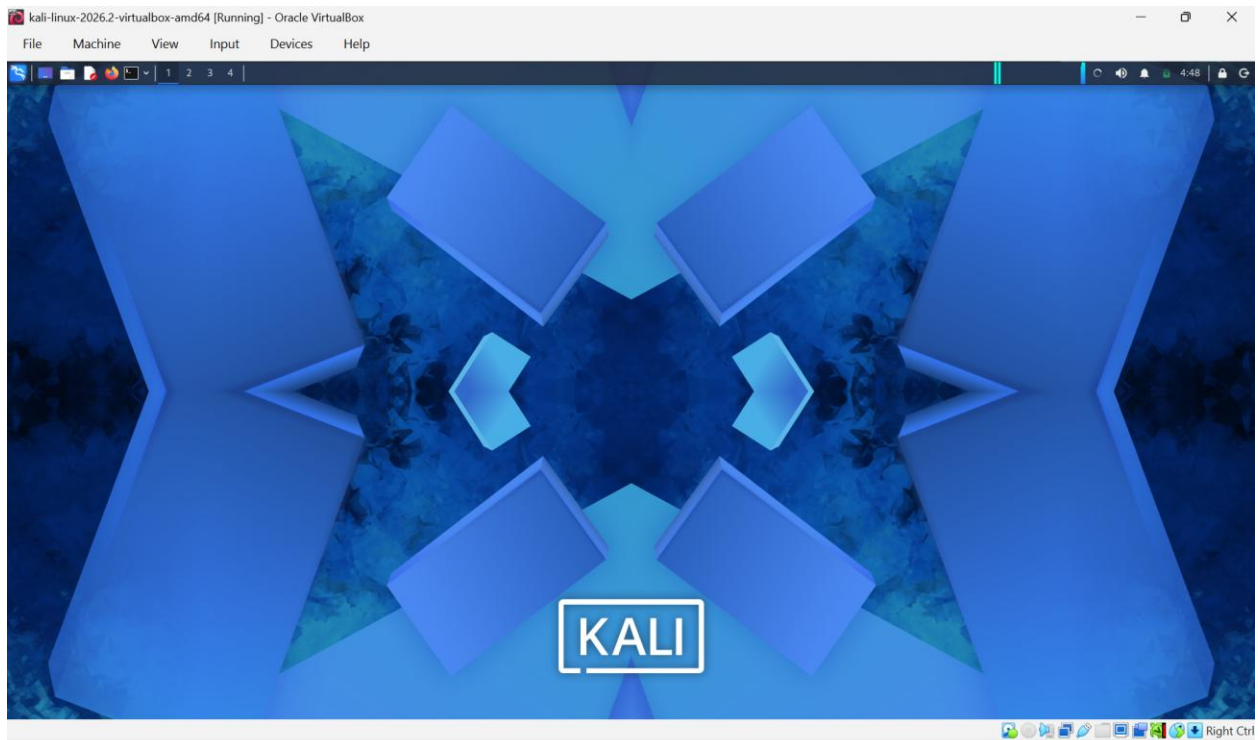


5.2.For metasploitable2:



2. Make sure both VMs boot cleanly and you can log in (Kali: your user; Metasploitable2: msfadmin/msfadmin).

For kali Linux:



For metasploitable2:

```

Login with msfadmin/msfadmin to get started

metasploitable login: * Reloading OpenBSD Secure Shell server's configuration
shd
...done.
* Reloading Postfix configuration...
...done.
msfadmin

Password:
Last login: Thu Jul 16 08:18:02 EDT 2026 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ _

```

Mode 1: NAT (Network Address Translation)

Configuration: Both Kali and Metasploitable2 set to NAT — the VirtualBox default.

Architecture:

Internet

|

Host (VirtualBox NAT Engine)

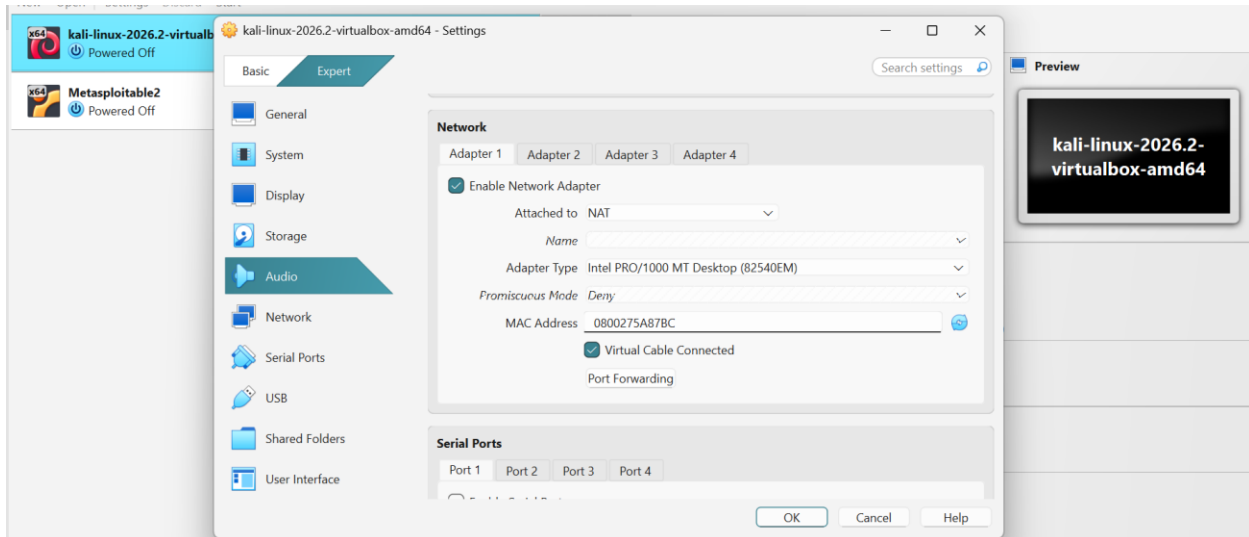
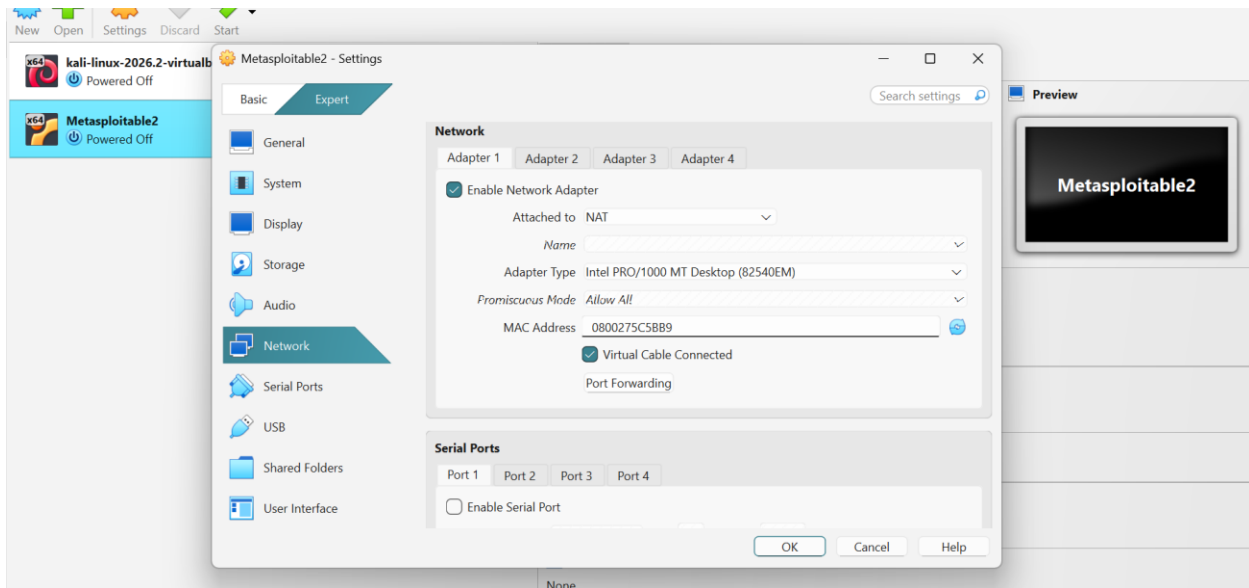
|

|— Kali (10.0.2.15) → [isolated private bubble]

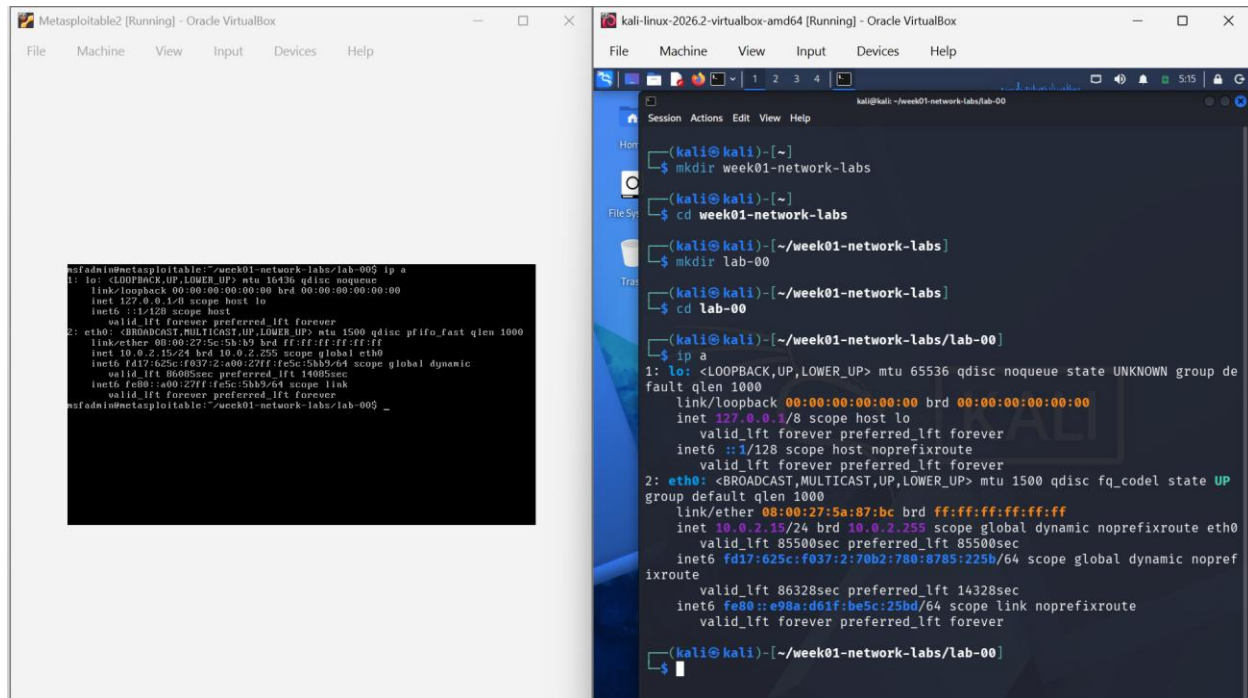
|

|— Metasploitable2 (10.0.2.15) → [separate isolated bubble]

No path between Kali and Metasploitable2



IP addressing check on both machines:



```
msfadmin@metasploitabl2:~/week01-network-labs/lab-00$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
link/ether 08:00:27:5c:5b:b9 brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global eth0
    inet6 fd17:625c:f037:2:a00:27ff:fe5c:5bb9/64 scope global dynamic
        valid_lft 86085sec preferred_lft 14005sec
    inet6 fe80::a00:27ff:fe5c:5bb9/64 scope link
        valid_lft forever preferred_lft forever
msfadmin@metasploitabl2:~/week01-network-labs/lab-00$ _

kali@kali:~/week01-network-labs/lab-00$
kali@kali:~/week01-network-labs$
kali@kali:~/week01-network-labs$
kali@kali:~/week01-network-labs$
kali@kali:~/week01-network-labs$
kali@kali:~/week01-network-labs/lab-00$
kali@kali:~/week01-network-labs/lab-00$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
    valid_lft 85500sec preferred_lft 85500sec
    inet6 fd17:625c:f037:2:70b2:780:8785:225b/64 scope global dynamic nopref
ixroute
        valid_lft 86328sec preferred_lft 14328sec
    inet6 fe80::e98a:d61f:be5c:25bd/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
kali@kali:~/week01-network-labs/lab-00$
```

For kali linux : 10.0.2.15/24 and broadcast address : 10.0.2.255

For metasploitabl2: 10.0.2.15/24 and broadcast address : 10.0.2.255

Both machines received the exact same IP address. This is not a configuration error — it is the defining behavior of NAT mode. VirtualBox creates a completely separate private virtual network for each VM individually. Each VM exists in its own isolated bubble, so two VMs can both be assigned 10.0.2.15 simultaneously without any conflict, because those addresses exist on different, unconnected network instances.

VM- TO -VM test

```
Metasploitable2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

msfadmin@metasploitable2:~/week01-network-labs/lab-00$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group de
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:5c:b9:bd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 85500sec preferred_lft 85500sec
    inet6 fd17:625c:f037:2:a00:27ff:fe5c:5b59/64 scope global dynamic nopre
        valid_lft 86328sec preferred_lft 14328sec
    link/ether 08:00:27:5c:b9:bd ff:ff:ff:ff:ff:ff
    inet6 fe80::e98a:d01f:be5c:25bd/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

msfadmin@metasploitable2:~/week01-network-labs/lab-00$ ping -c 4 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data:
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=0.623 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=0.070 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=0.002 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=0.090 ms

--- 10.0.2.15 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2990ms
rtt min/avg/max/mdev = 0.070/0.220/0.623/0.232 ms
msfadmin@metasploitable2:~/week01-network-labs/lab-00$

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali:~/week01-network-labs/lab-00$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group de
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 85500sec preferred_lft 85500sec
    inet6 fd17:625c:f037:2:70b2:780:8785:225b/64 scope global dynamic nopre
        valid_lft 86328sec preferred_lft 14328sec
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet6 fe80::e98a:d01f:be5c:25bd/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

kali@kali:~/week01-network-labs/lab-00$ ping -c 4 10.0.2.15
ping: invalid argument: '10.0.2.15'

kali@kali:~/week01-network-labs/lab-00$ ping -c 4 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data:
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=0.269 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=0.047 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=0.049 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=0.044 ms

--- 10.0.2.15 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3057ms
rtt min/avg/max/mdev = 0.044/0.102/0.269/0.096 ms

kali@kali:~/week01-network-labs/lab-00$ ping -c 4 10.0.2.15
ping: invalid argument: '10.0.2.15'

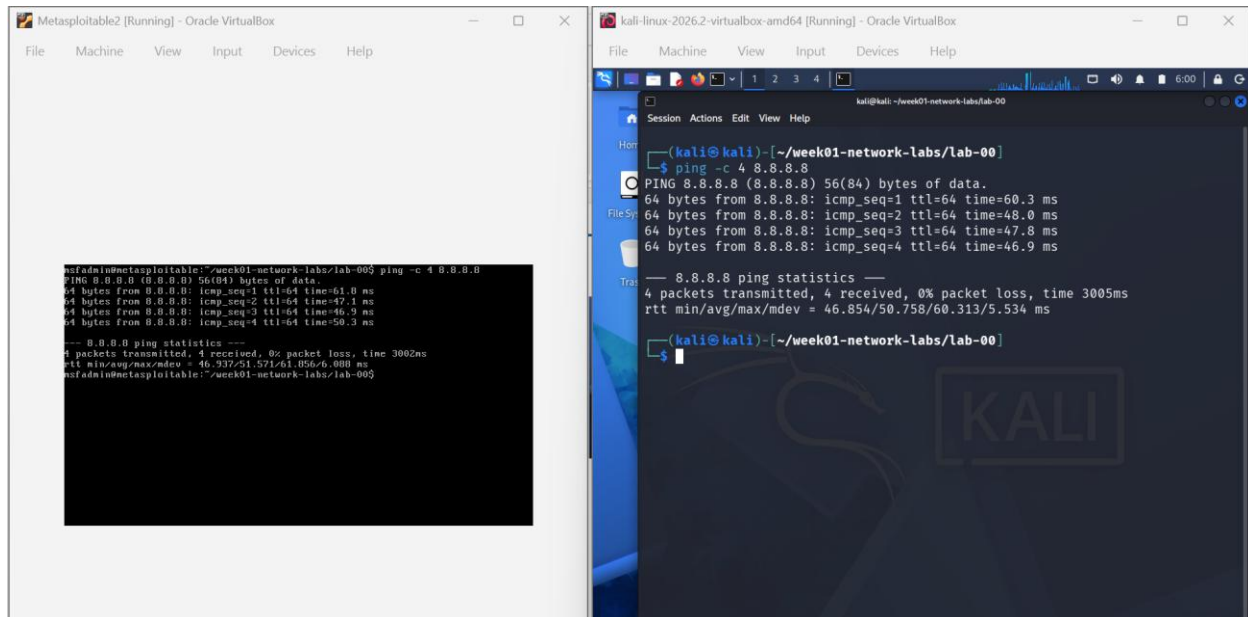
kali@kali:~/week01-network-labs/lab-00$ ping -c 4 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data:
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=0.269 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=0.047 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=0.049 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=0.044 ms

--- 10.0.2.15 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3057ms
rtt min/avg/max/mdev = 0.044/0.102/0.269/0.096 ms

kali@kali:~/week01-network-labs/lab-00$
```

When Kali pings 10.0.2.15, it is pinging itself its own IP, on its own isolated network. There is no other device on Kali's NAT network. The ping "succeeds" but the sub-millisecond response time is the giveaway: the packet never left Kali's own networking stack. This is a self-loop, not genuine VM-to-VM communication.

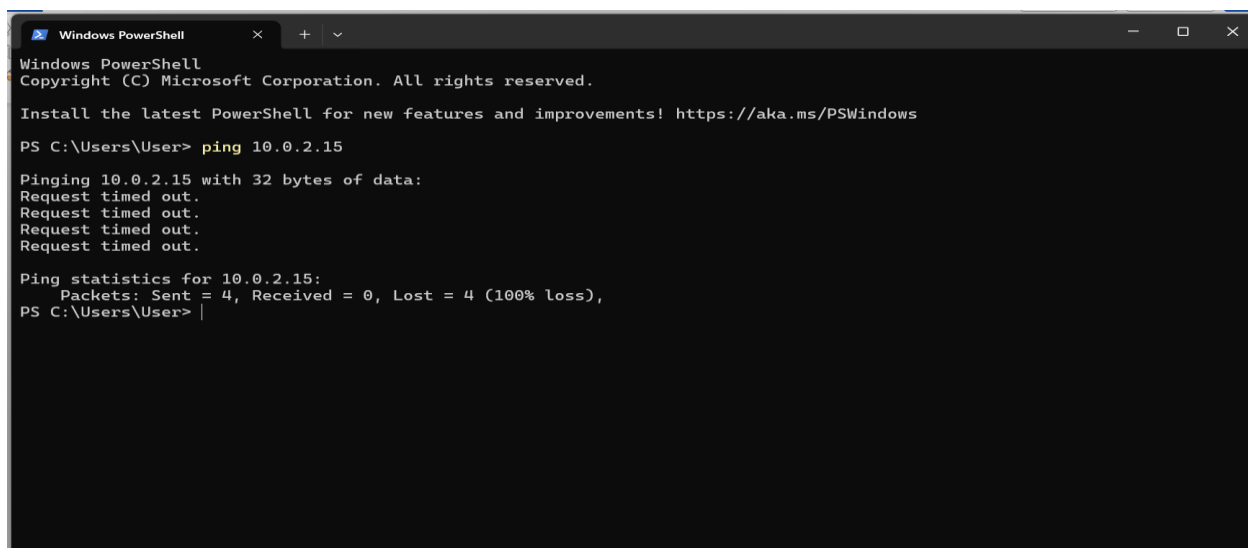
Internet test:



Both VMs successfully reached 8.8.8.8. VirtualBox's NAT engine intercepts all outbound traffic from each VM, rewrites the source address to the host's real IP, and forwards it through the host's actual network connection. Replies come back to VirtualBox, which forwards them to the correct VM. The VM never knows its traffic is being translated — this is how NAT works in real routers as well (your home router does the same thing for all devices on your WiFi).

Host-to-VM test:

Kali ping ip address on powershell:



Metasploitable2 ping ip address on powershell:

```

Ping statistics for 10.0.2.15:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PS C:\Users\User> ping 10.0.2.15

Pinging 10.0.2.15 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.2.15:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PS C:\Users\User> |

```

Not applicable for NAT mode. The host has no route into the VM's isolated private network. VirtualBox's NAT engine only translates outbound traffic from VMs — it does not provide a path inward from the host.

Why NAT mode exists: It is the safest and simplest default. A VM gets internet access immediately without any configuration, and it is completely unreachable from the outside. Suitable for a single VM that needs internet access and does not need to communicate with other VMs.

Why it fails for security labs: Any lab requiring two machines to interact (attacker → target) is impossible in NAT mode. The machines are on separate networks and cannot discover or reach each other.

Mode 2: NAT Network

Configuration: Both VMs set to NAT Network. A NAT Network named "NatNetwork" was created first via VirtualBox Network Manager (File → Tools → Network Manager → NAT Networks → Create).

Architecture:

Internet

|

Host (VirtualBox NAT Engine)

|

NAT Network (10.0.2.0/24) — one shared network

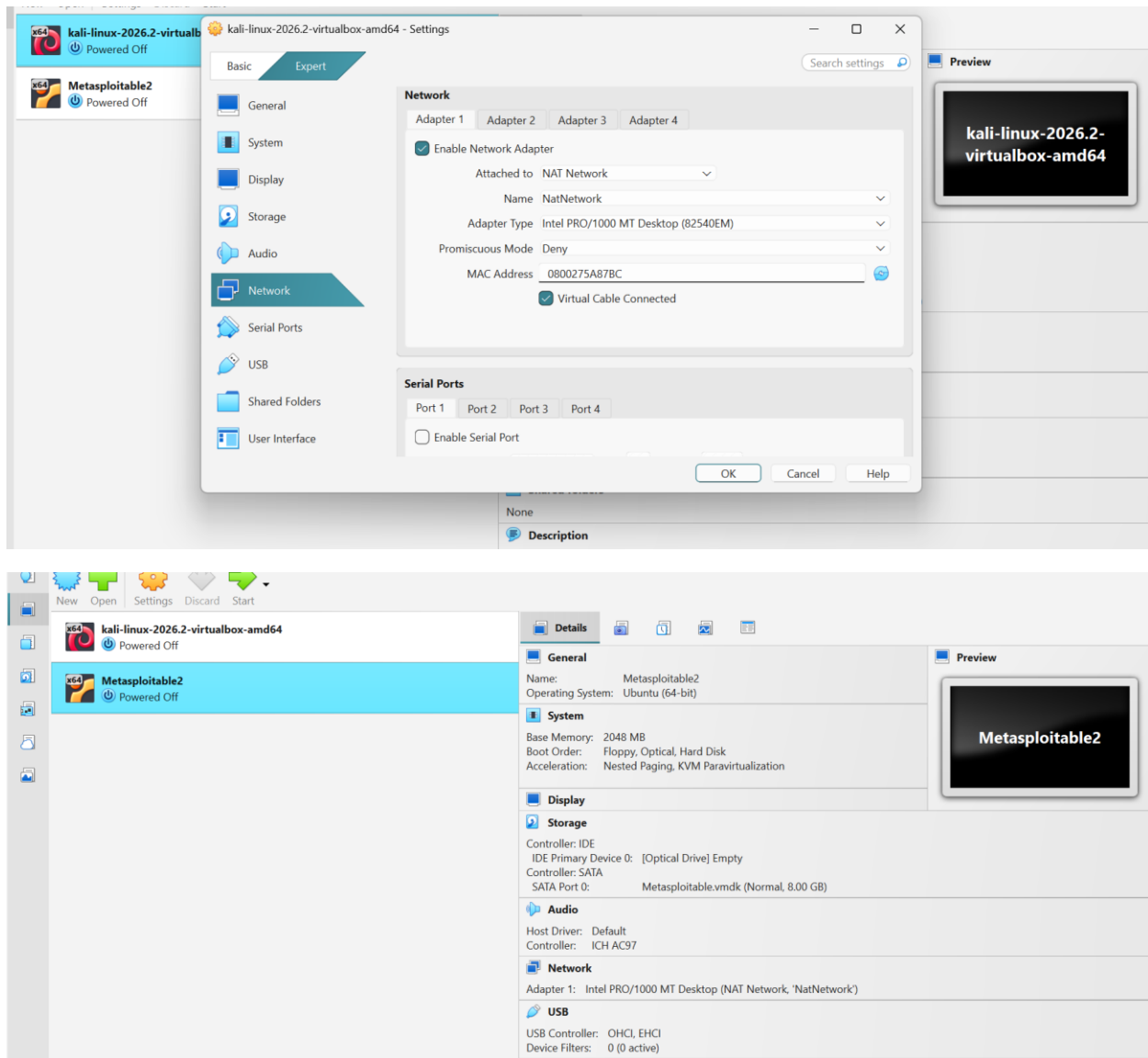
|

├── Kali (10.0.2.15)

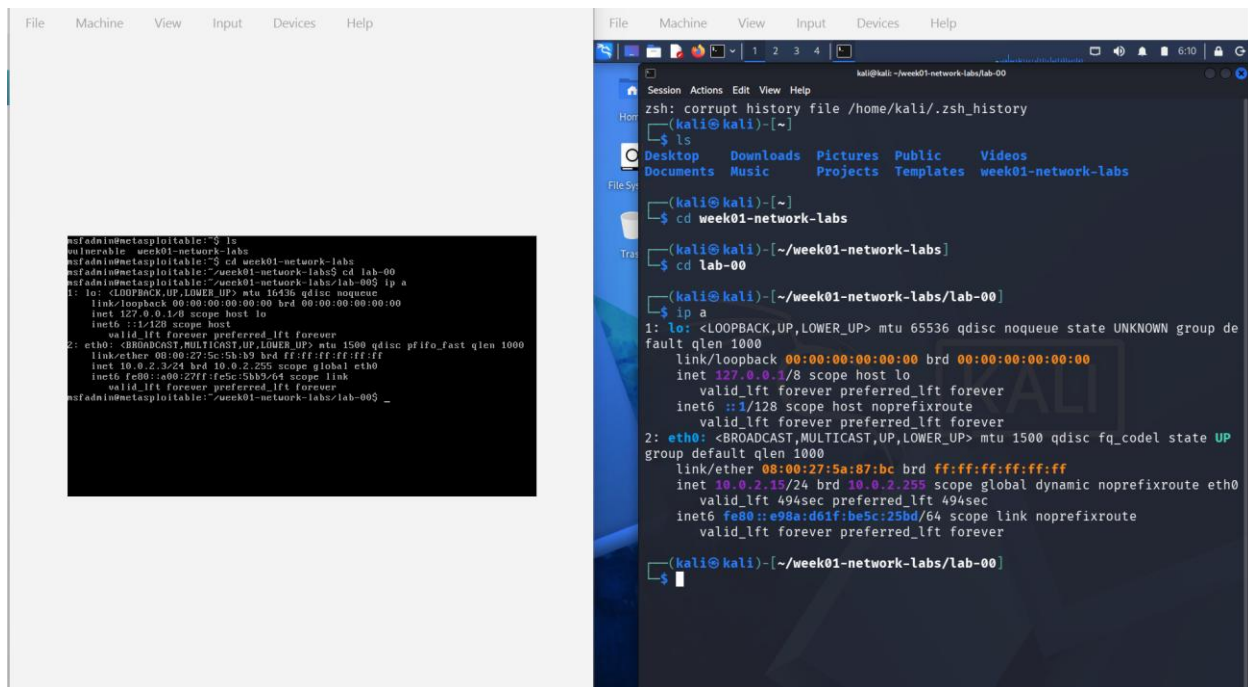
|

└── Metasploitable2 (10.0.2.3)

Both VMs share one network: VM-to-VM communication + internet



IP addresses:

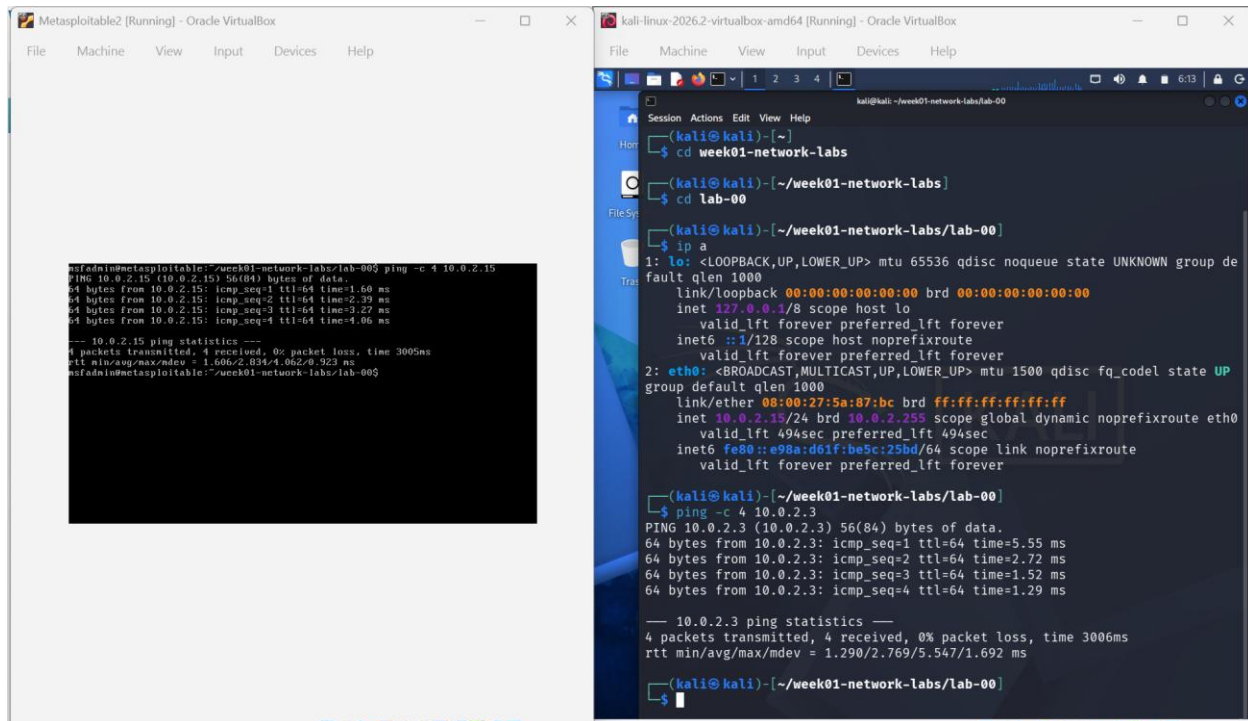


For kali linux: inet :10.0.2.15/24 broadcast address: 10.0.2.255

For metasploitable2: inet: 10.0.2.3/24 broadcast address: 10.0.2.255

This time the addresses are different, both within the same subnet. This is because NAT Network places all VMs on a single shared virtual network with one common DHCP server. Both machines drew from the same address pool, so they received different addresses — unlike NAT mode where each had its own isolated pool and both ended up at .15.

VM-to-VM test:



```
msfadmin@metasploitable:~/week01-network-labs/lab-00$ ping -c 4 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data:
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=1.50 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=2.39 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=3.27 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=4.06 ms

--- 10.0.2.15 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.006/2.834/4.062/0.923 ms
msfadmin@metasploitable:~/week01-network-labs/lab-00$

(kali@kali)-[~/week01-network-labs]
$ cd lab-00

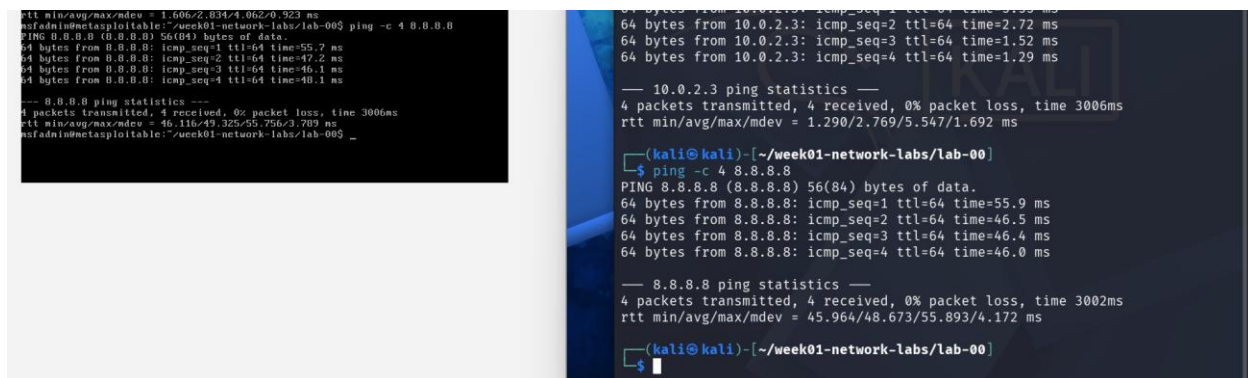
(kali@kali)-[~/week01-network-labs/lab-00]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group de
fault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
group default qlen 1000
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 494sec preferred_lft 494sec
    inet6 fe80::e98a:d01f:be5c:25bd/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)-[~/week01-network-labs/lab-00]
$ ping -c 4 10.0.2.3
PING 10.0.2.3 (10.0.2.3) 56(84) bytes of data:
64 bytes from 10.0.2.3: icmp_seq=1 ttl=64 time=5.55 ms
64 bytes from 10.0.2.3: icmp_seq=2 ttl=64 time=2.72 ms
64 bytes from 10.0.2.3: icmp_seq=3 ttl=64 time=1.52 ms
64 bytes from 10.0.2.3: icmp_seq=4 ttl=64 time=1.29 ms

--- 10.0.2.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.290/2.769/5.547/1.692 ms

(kali@kali)-[~/week01-network-labs/lab-00]
$
```

Test internet:



```
rtt min/avg/max/mdev = 1.606/2.834/4.062/0.923 ms
msfadmin@metasploitable:~/week01-network-labs/lab-00$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=55.7 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=47.2 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=46.1 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=48.1 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 46.116/49.325/55.756/3.789 ms
msfadmin@metasploitable:~/week01-network-labs/lab-00$

(kali@kali)-[~/week01-network-labs/lab-00]
$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=55.9 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=46.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=46.4 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=46.0 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3002ms
rtt min/avg/max/mdev = 45.964/48.673/55.893/4.172 ms

(kali@kali)-[~/week01-network-labs/lab-00]
$
```



```
Metasploitable2 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

msfadmin@metasploitable1: /week01-network-labs/lab-005$ ping -c 4 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data:
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=1.60 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=2.39 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=3.27 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=4.06 ms

--- 10.0.2.15 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.506/2.834/4.062/0.923 ms
msfadmin@metasploitable1: /week01-network-labs/lab-005$ ping -c 4 0.0.0.0
PING 0.0.0.0 (0.0.0.0) 56(84) bytes of data:
64 bytes from 0.0.0.0: icmp_seq=1 ttl=64 time=55.7 ms
64 bytes from 0.0.0.0: icmp_seq=2 ttl=64 time=47.2 ms
64 bytes from 0.0.0.0: icmp_seq=3 ttl=64 time=46.1 ms
64 bytes from 0.0.0.0: icmp_seq=4 ttl=64 time=40.1 ms

--- 0.0.0.0 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 46.116/49.325/55.756/3.789 ms
msfadmin@metasploitable1: /week01-network-labs/lab-005_

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

kali@kali: /week01-network-labs/lab-00$
Session Actions Edit View Help
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valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
group default qlen 1000
link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
valid_lft 494sec preferred_lft 494sec
inet6 fe80::e98a:d61f:be5c:25bd/64 scope link noprefixroute
valid_lft forever preferred_lft forever

(kali@kali)-[~/week01-network-labs/lab-00]
$ ping -c 4 10.0.2.3
PING 10.0.2.3 (10.0.2.3) 56(84) bytes of data:
64 bytes from 10.0.2.3: icmp_seq=1 ttl=64 time=5.55 ms
64 bytes from 10.0.2.3: icmp_seq=2 ttl=64 time=2.72 ms
64 bytes from 10.0.2.3: icmp_seq=3 ttl=64 time=1.52 ms
64 bytes from 10.0.2.3: icmp_seq=4 ttl=64 time=1.29 ms

--- 10.0.2.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.290/2.769/5.547/1.692 ms

(kali@kali)-[~/week01-network-labs/lab-00]
$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=55.9 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=46.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=46.4 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=46.0 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3002ms
rtt min/avg/max/mdev = 45.964/48.673/55.893/4.172 ms

(kali@kali)-[~/week01-network-labs/lab-00]
```

Communication succeeded in both directions. The latency (2–9ms) is measurably higher than the sub-millisecond self-loop seen in NAT mode this confirms genuine packets were traversing a network path between two separate machines, not looping back

Test host to VM:

```
Windows PowerShell
Pinging 10.0.2.15 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.2.15:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PS C:\Users\User> ping 10.0.2.15

Pinging 10.0.2.15 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.2.15:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PS C:\Users\User> ping 10.0.2.3

Pinging 10.0.2.3 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.2.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PS C:\Users\User>
```

Not directly applicable. The host is not a member of the NAT Network's shared subnet.

Why NAT Network is better than NAT for labs: It solves NAT's core limitation by putting multiple VMs on the same network. VMs can discover and reach each other while still having internet access. Suitable for labs that need both VM-to-VM communication and internet (e.g. to pull tools, updates, or run network scans that require name resolution).

Mode 3: Host only

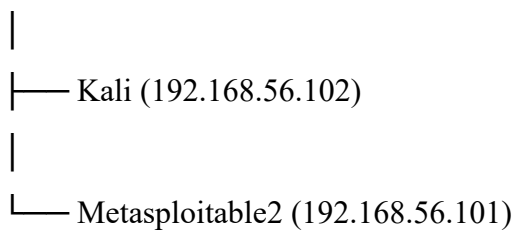
Configuration: Both VMs set to Host-Only Adapter using the VirtualBox Host-Only Ethernet Adapter (192.168.56.0/24), which has its own built-in DHCP server.

Architecture:

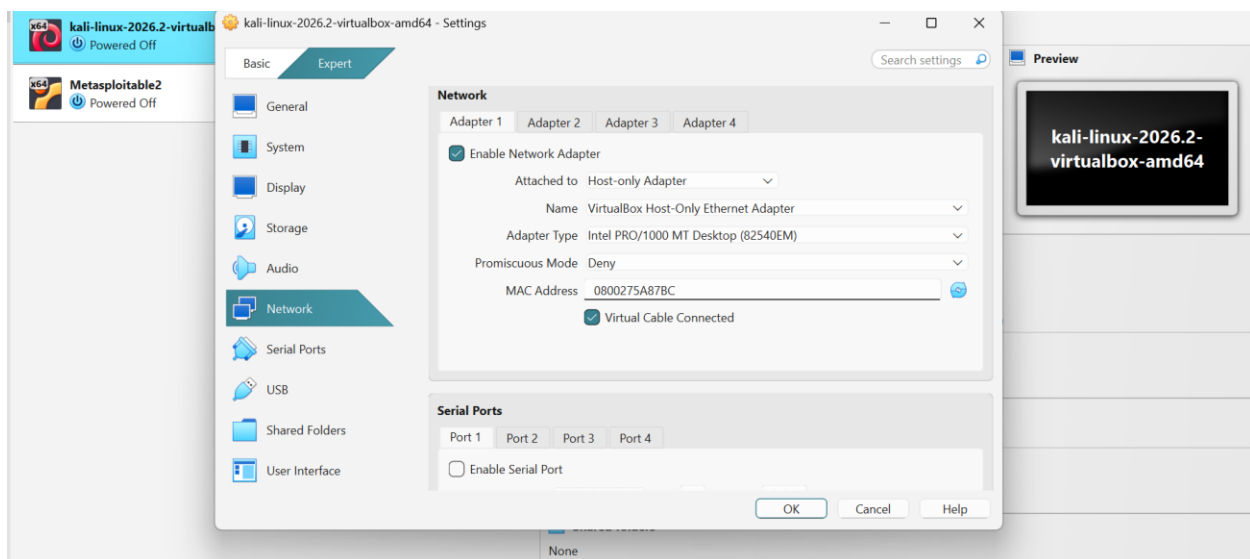
Internet

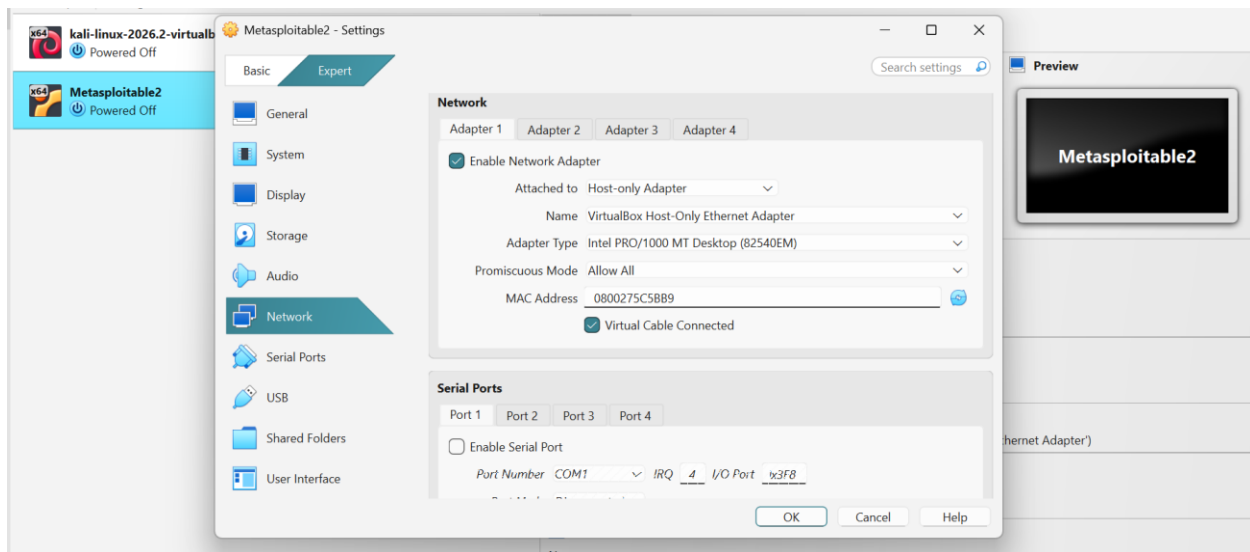
✗ (no route out)

Host — Host-Only Network (192.168.56.0/24)

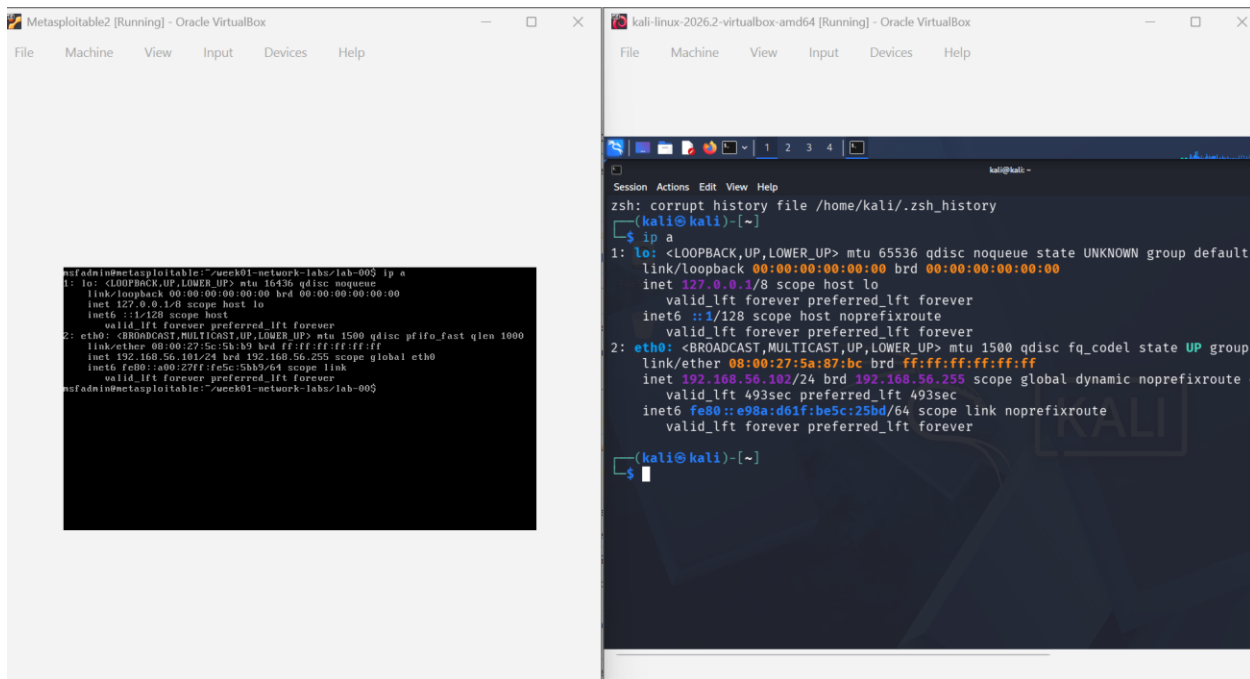


Host can reach both VMs. No internet access.





IP addresses:



For kali linux: inet :192.168.56.102/24 broadcast address: 192.168.56.255

For metasploitable2: inet: 192.168.56.101/24 broadcast address: 192.168.56.255

Both addresses were assigned automatically by the Host-Only network's built-in DHCP server — no manual configuration needed.

VM-to-VM test:

```
nsfadmin@metasploitkali:~/work01-network-labs/lab-005$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16384 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        inet6 ::1/128 scope host
            valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:5c:5b:b9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global eth0
        inet6 fe80::e98a:d61f:be5c:25bd/64 scope link
            valid_lft forever preferred_lft forever
nsfadmin@metasploitkali:~/work01-network-labs/lab-005$ ping -c 4 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=1.42 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=1.94 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=2.45 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=2.13 ms

--- 192.168.56.102 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3002ms
rtt min/avg/max/mdev = 1.427/1.980/2.451/0.371 ms
nsfadmin@metasploitkali:~/work01-network-labs/lab-005$
```

```
kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute
        valid_lft 493sec preferred_lft 493sec
    inet6 fe80::e98a:d61f:be5c:25bd/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=5.58 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.35 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=3.23 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=1.28 ms

--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.276/2.857/5.580/1.755 ms

(kali@kali)-[~]
$
```

```
nsfadmin@metasploitkali:~/work01-network-labs/lab-005$ ping -c 4 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=1.42 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=1.94 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=2.45 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=2.13 ms

--- 192.168.56.102 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3002ms
rtt min/avg/max/mdev = 1.427/1.980/2.451/0.371 ms
nsfadmin@metasploitkali:~/work01-network-labs/lab-005$
```

```
inet6 fe80::e98a:d61f:be5c:25bd/64 scope link noprefixroute
    valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=5.58 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.35 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=3.23 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=1.28 ms

--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.276/2.857/5.580/1.755 ms

(kali@kali)-[~]
$
```

Succeeded in both directions. Both machines are on the same 192.168.56.0/24 subnet, so they can communicate directly.

Internet test:

```
nsfadmin@metasploitkali:~/work01-network-labs/lab-005$ ping -c 4 8.8.8.8
connect: Network is unreachable
nsfadmin@metasploitkali:~/work01-network-labs/lab-005$
```

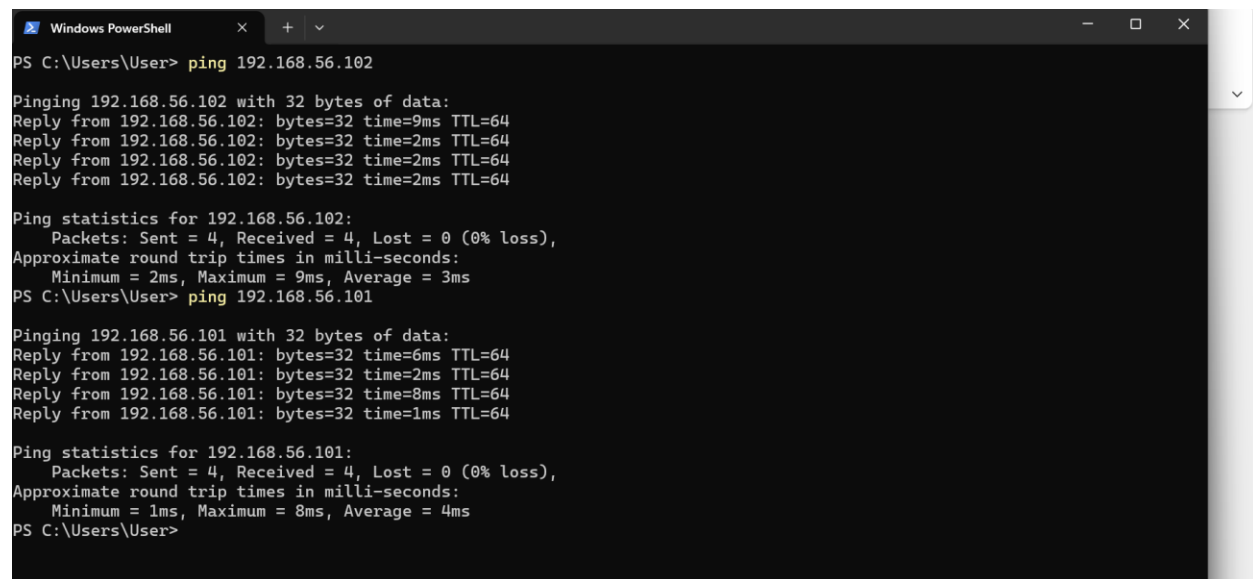
```
kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
$ ping -c 4 8.8.8.8
ping: connect: Network is unreachable

(kali@kali)-[~]
$
```

Failed on both VMs with "Network is unreachable". The Host-Only network is entirely self-contained. VirtualBox provides no outbound path to the internet or the physical network. The kernel's routing table has no default gateway to send packets to, so it rejects them before they ever hit the wire.

Host to VM test:



```
Windows PowerShell
PS C:\Users\User> ping 192.168.56.102

Pinging 192.168.56.102 with 32 bytes of data:
Reply from 192.168.56.102: bytes=32 time=9ms TTL=64
Reply from 192.168.56.102: bytes=32 time=2ms TTL=64
Reply from 192.168.56.102: bytes=32 time=2ms TTL=64
Reply from 192.168.56.102: bytes=32 time=2ms TTL=64

Ping statistics for 192.168.56.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 9ms, Average = 3ms
PS C:\Users\User> ping 192.168.56.101

Pinging 192.168.56.101 with 32 bytes of data:
Reply from 192.168.56.101: bytes=32 time=6ms TTL=64
Reply from 192.168.56.101: bytes=32 time=2ms TTL=64
Reply from 192.168.56.101: bytes=32 time=8ms TTL=64
Reply from 192.168.56.101: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.56.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 8ms, Average = 4ms
PS C:\Users\User>
```

Succeeded from Windows PowerShell to both VMs. This is the defining characteristic of Host-Only mode the Windows host itself has a virtual interface connected to the Host-Only network (you can see it in Windows network adapters as "VirtualBox Host-Only Ethernet Adapter"), and through this interface it can reach all VMs on that network directly.

Why this matters for labs: Host-Only is ideal when you want to run tools from your Windows host machine against the VMs (e.g. connecting to a web service on Metasploitable2 from your browser, or SSH-ing into Kali from your terminal). The host can reach everything, but the vulnerable VM cannot reach the internet and cannot be reached from outside.

Mode 4: Internal Network

Configuration: Both VMs set to Internal Network with the same network name (intnet). No DHCP server exists in this mode static IPs were manually assigned using:

```
sudo ifconfig eth0 192.168.56.10 netmask 255.255.255.0 up (on Kali)
```

`sudo ifconfig eth0 192.168.56.20 netmask 255.255.255.0 up` (on Metasploitable2)

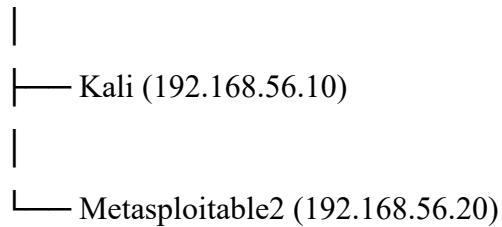
Architecture:

Internet

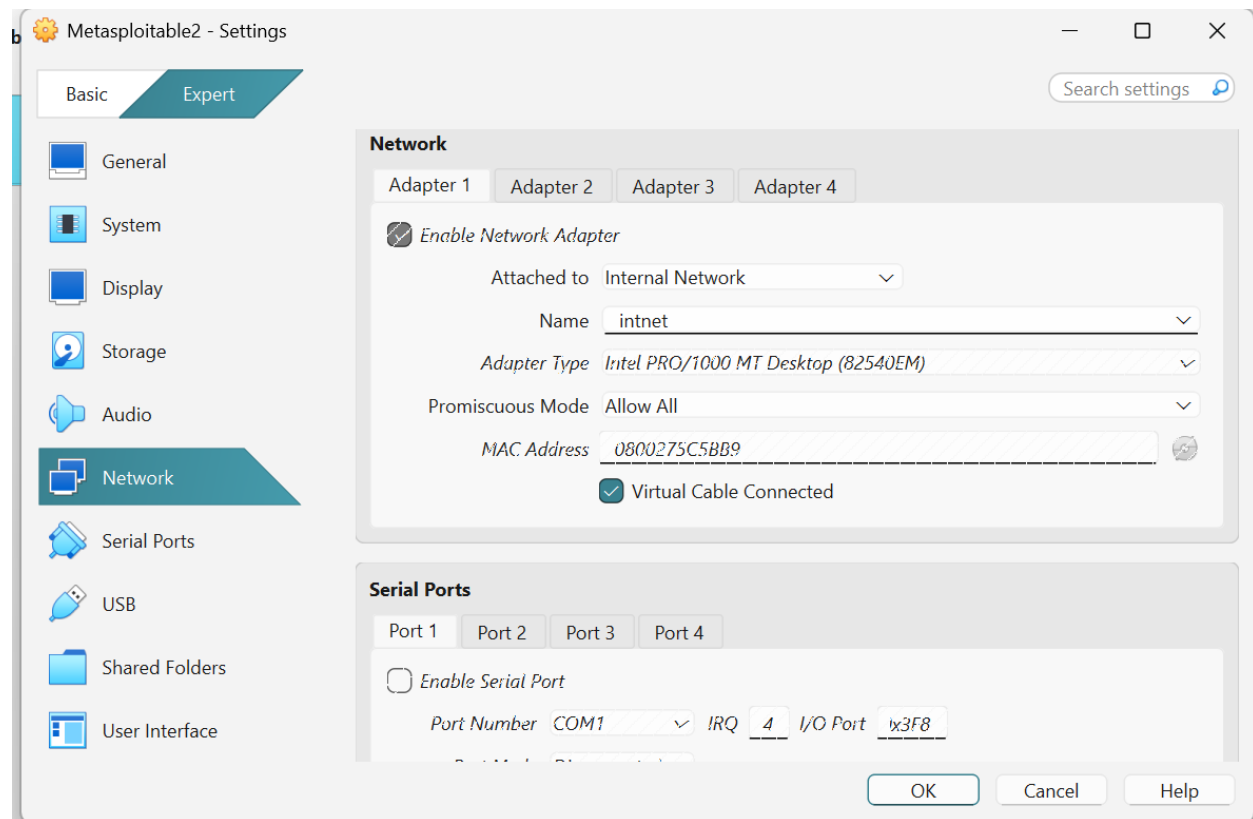
✗ (no route)

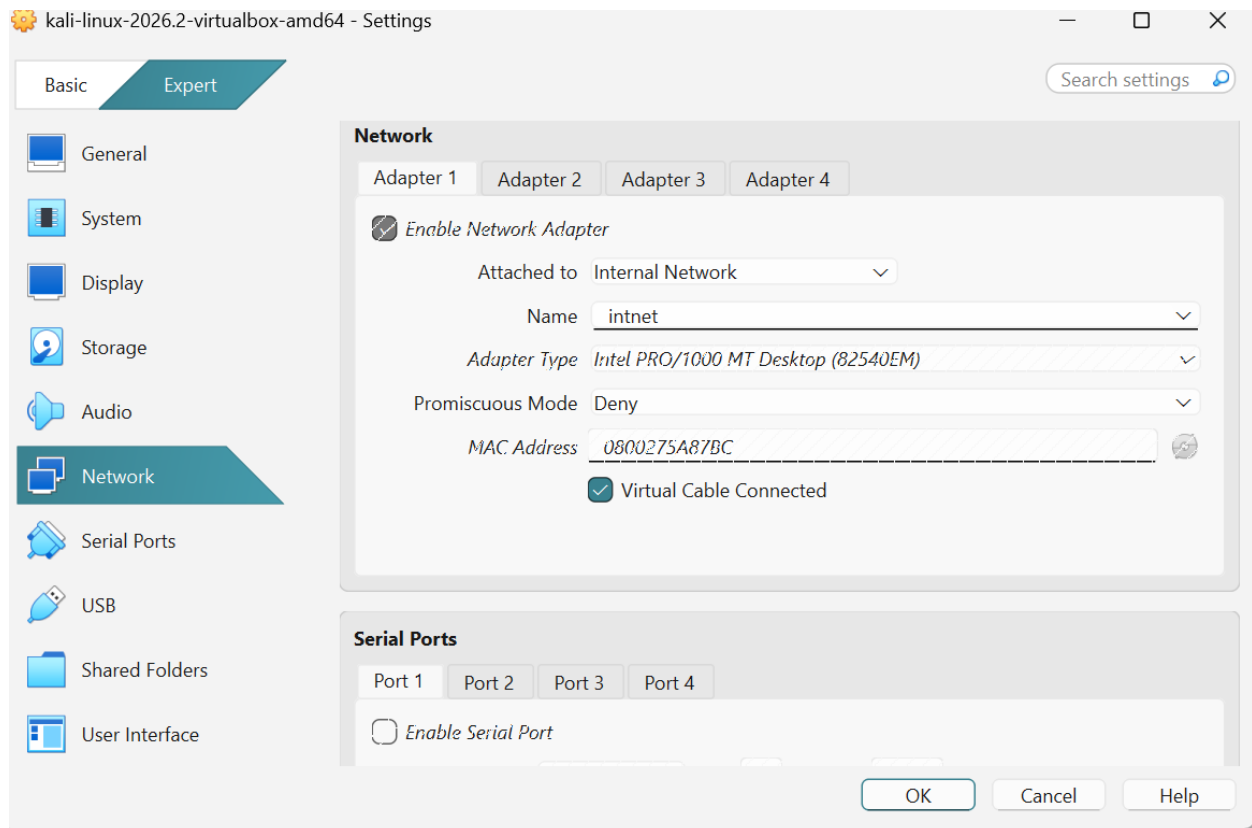
Host

✗ (no route) Internal Network (intnet)

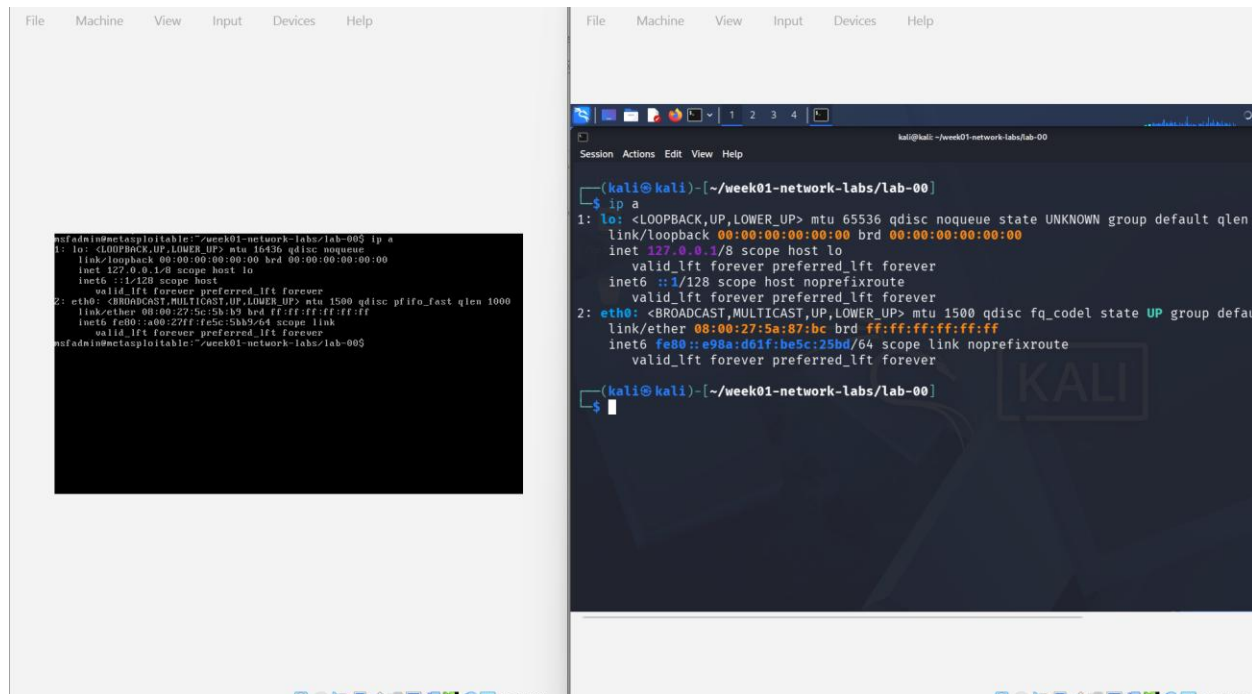


VM-to-VM only. Host and internet completely excluded.





IP addresses:




```

msfadmin@metasploitable:~/week01-network-labs/lab-00$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16384 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        inet6 ::1/128 scope host
            valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 00:00:27:5e:5b:b9 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::a00:27ff:fe5b:b9/64 scope link
        valid_lft forever preferred_lft forever
msfadmin@metasploitable:~/week01-network-labs/lab-00$ sudo ifconfig eth0 192.168.56.20 netmask 255.255.0.0 up
[sudo] password for msfadmin:
msfadmin@metasploitable:~/week01-network-labs/lab-00$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16384 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        inet6 ::1/128 scope host
            valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 00:00:27:5e:5b:b9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.20/24 brd 192.168.56.255 scope global eth0
        inet6 fe80::a00:27ff:fe5b:b9/64 scope link
            valid_lft forever preferred_lft forever
msfadmin@metasploitable:~/week01-network-labs/lab-00$

```

```

kali@kali: ~/week01-network-labs/lab-00
Session Actions Edit View Help

link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet6 fe80::e98a:d61f:be5c:25bd/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali) - [~/week01-network-labs/lab-00]
$ sudo ifconfig eth0 192.168.56.10 netmask 255.255.0.0 up
[sudo] password for kali:

(kali@kali) - [~/week01-network-labs/lab-00]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.10/24 brd 192.168.56.255 scope global eth0
        valid_lft forever preferred_lft forever

(kali@kali) - [~/week01-network-labs/lab-00]

```

For kali linux: inet:192.168.56.10 broadcast address:192.168.56.255

For metasploitable2: inet:192.168.56.20 broadcast address:192.168.56.255

When both VMs first booted with no IP configured, Metasploitable2 showed no IPv4 address at all on eth0 — only the link-local address. This is expected: Internal Network has no DHCP server by design. Static assignment is mandatory.

VM-to-VM test:

```

msfadmin@metasploitable:~/week01-network-labs/lab-00$ ping -c 4 192.168.56.10
PING 192.168.56.10 (192.168.56.10) 56(84) bytes of data:
64 bytes from 192.168.56.10: icmp_seq=1 ttl=64 time=20.4 ms
64 bytes from 192.168.56.10: icmp_seq=2 ttl=64 time=2.64 ms
64 bytes from 192.168.56.10: icmp_seq=3 ttl=64 time=3.90 ms
64 bytes from 192.168.56.10: icmp_seq=4 ttl=64 time=2.66 ms

--- 192.168.56.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 301ms
rtt min/avg/max/mdev = 2.640/2.460/20.426/7.531 ms
msfadmin@metasploitable:~/week01-network-labs/lab-00$

```

```

kali@kali: ~/week01-network-labs/lab-00
Session Actions Edit View Help

(kali@kali) - [~/week01-network-labs/lab-00]
$ ping -c 4 192.168.56.20
PING 192.168.56.20 (192.168.56.20) 56(84) bytes of data:
64 bytes from 192.168.56.20: icmp_seq=1 ttl=64 time=3.12 ms
64 bytes from 192.168.56.20: icmp_seq=2 ttl=64 time=1.64 ms
64 bytes from 192.168.56.20: icmp_seq=3 ttl=64 time=2.81 ms
64 bytes from 192.168.56.20: icmp_seq=4 ttl=64 time=2.14 ms

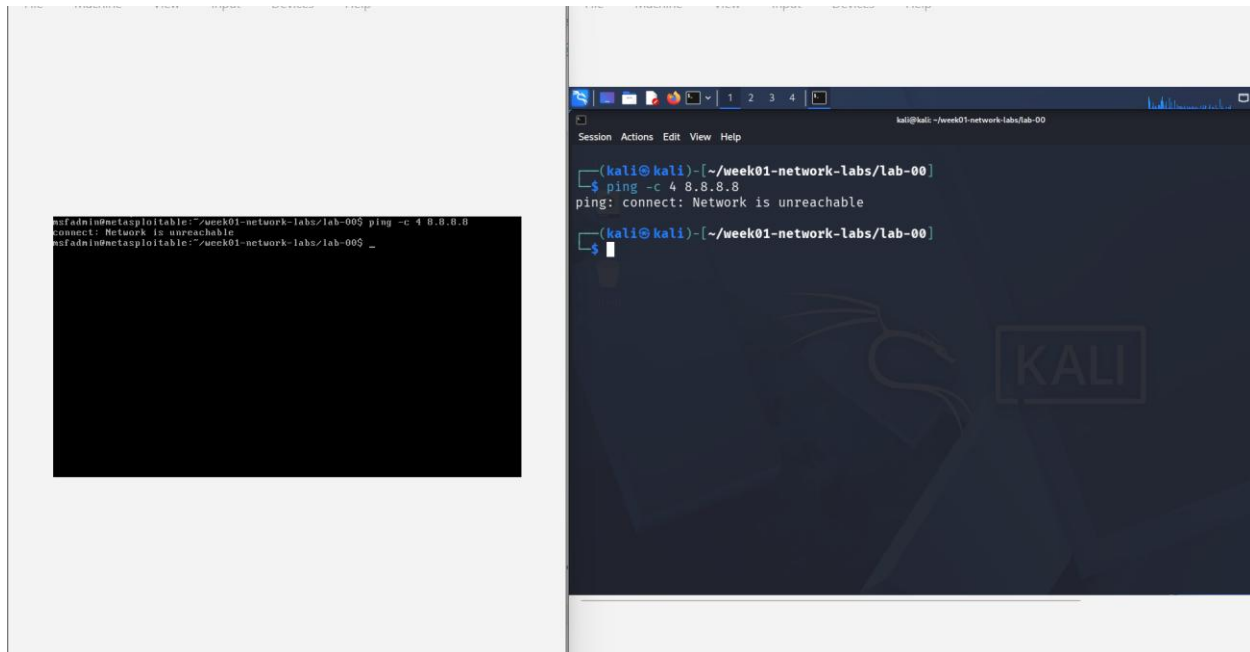
--- 192.168.56.20 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 302ms
rtt min/avg/max/mdev = 1.644/2.427/3.123/0.575 ms

(kali@kali) - [~/week01-network-labs/lab-00]

```

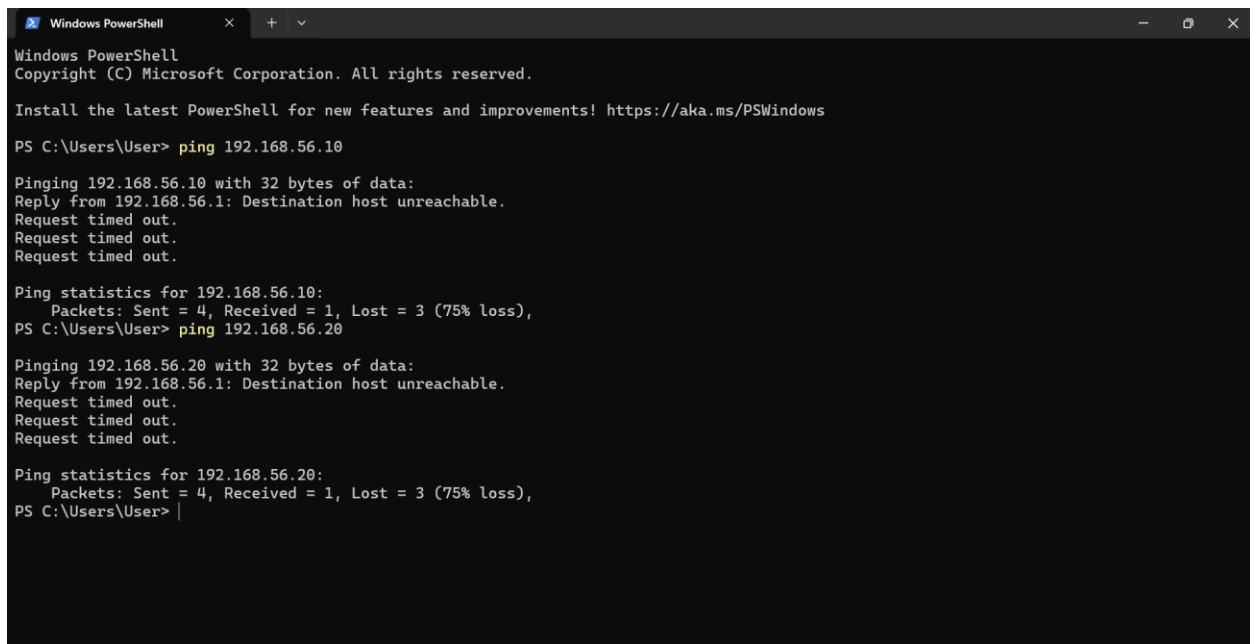

Succeeded after manual IP assignment. Both VMs are on the same subnet and can communicate directly.

Internet test:



Failed with "Network is unreachable" same reason as Host-Only, no gateway exists.

Host to VM test:



Failed with "Destination host unreachable" from PowerShell. The error text is slightly different from the VM-side error and the distinction is important: when Windows tried to ping

192.168.56.10, its own routing table had no idea how to reach that subnet. The ping request was forwarded to Windows' default gateway (your real router), which responded with "Destination host unreachable" because no such device exists on the real network. The VM never received the ping at all. This is completely different from Host-Only, where the host has a dedicated virtual interface into the lab network.

Why Internal Network is the strictest mode: Nothing can enter or exit except between VMs on the same named internal network. This is the most appropriate mode for controlled security experiments where you want zero risk of traffic leaking to or from the real network. The trade-off is manual IP management and no host access.

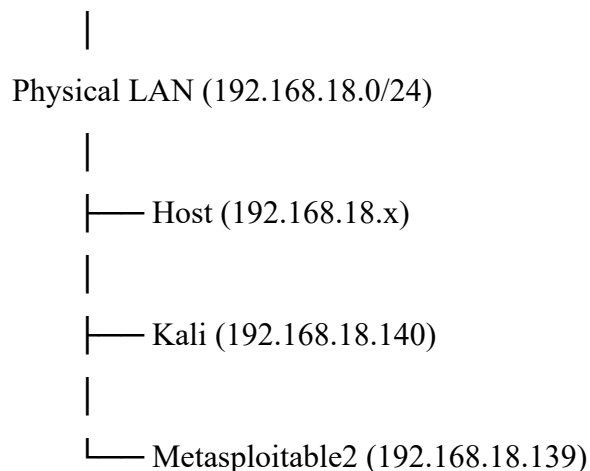
Why we use this mode for the packet capture, ARP, and scanning labs: It gives completely clean traffic every packet you see in Wireshark is yours, generated by your lab. No background noise from the host's real network.

Mode 5: Bridged

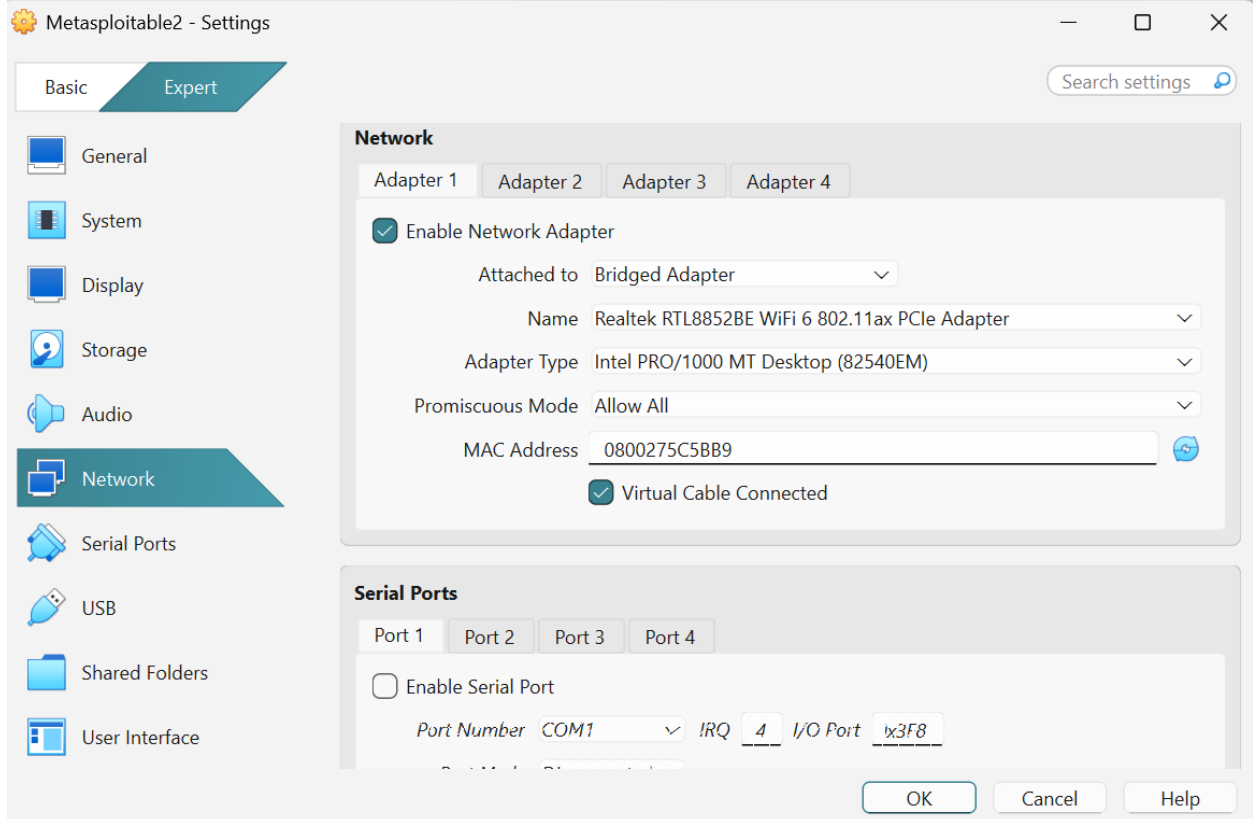
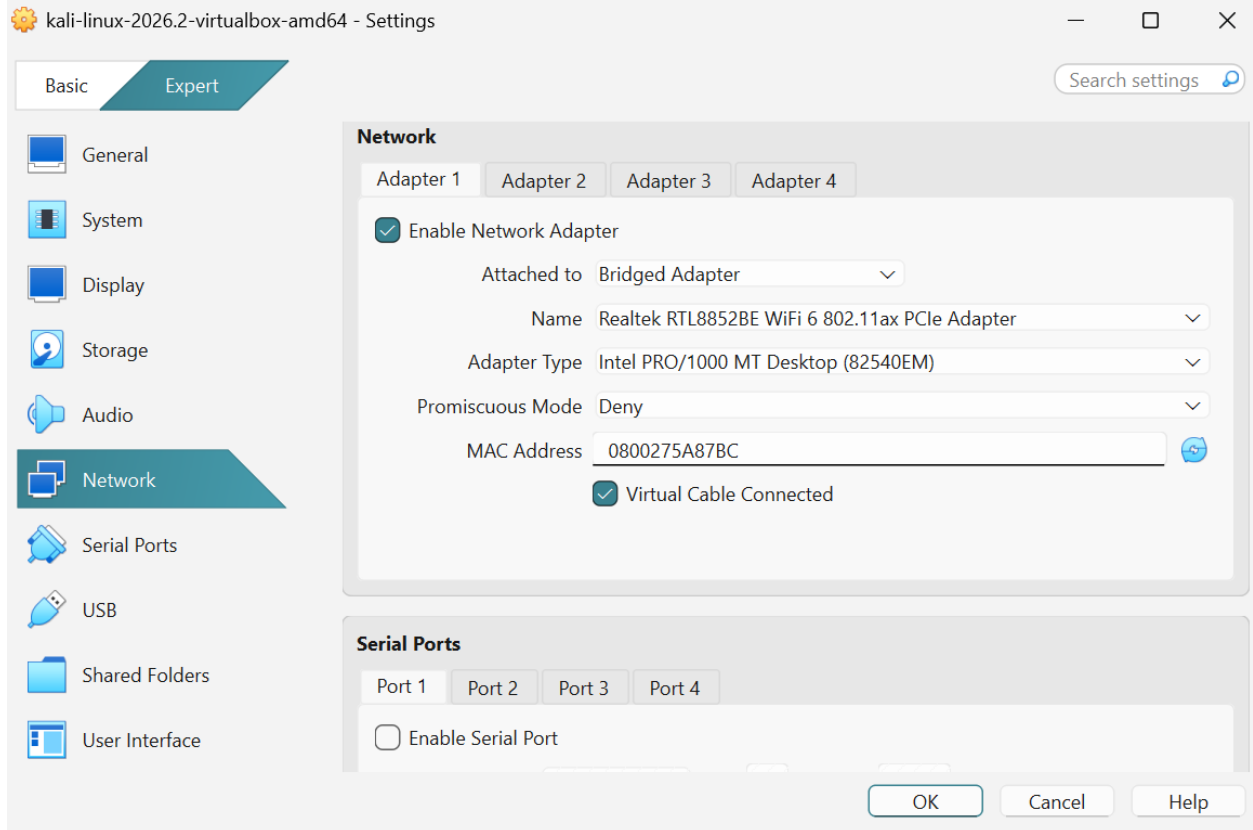
Configuration: Both VMs set to Bridged Adapter, connected to the host's physical WiFi adapter (Realtek RTL8852BE).

Architecture:

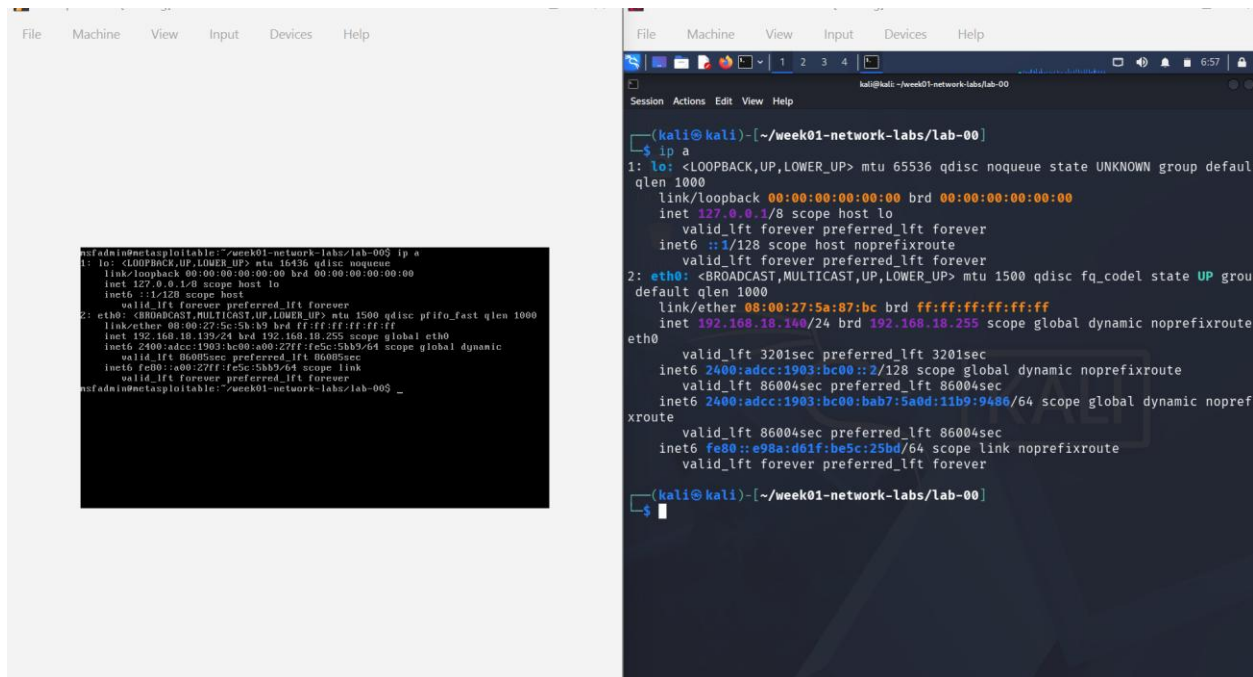
Physical Router / Internet



All devices are full peers on the real network



IP addresses:



The left terminal shows the configuration of the Kali Linux VM's network interface 'lo' and 'eth0'. The 'lo' interface is assigned the IP 127.0.0.1. The 'eth0' interface is assigned the IP 192.168.18.140. The right terminal shows the configuration of the Metasploitable2 VM's network interface 'eth0', which is assigned the IP 192.168.18.139.

```
(kali@kali)-[~/week01-network-labs/lab-00]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        inet6 ::1/128 scope host
            valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:5c:5b:b9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.18.140/24 brd 192.168.18.255 scope global dynamic eth0
        inet6 2400:adcc:1903:bc00:a00:27ff:fe5c:5bb9/64 scope global dynamic
            valid_lft 86005sec preferred_lft 86005sec
        inet6 fe80::a00:27ff:fe5c:5bb9/64 scope link
            valid_lft forever preferred_lft forever

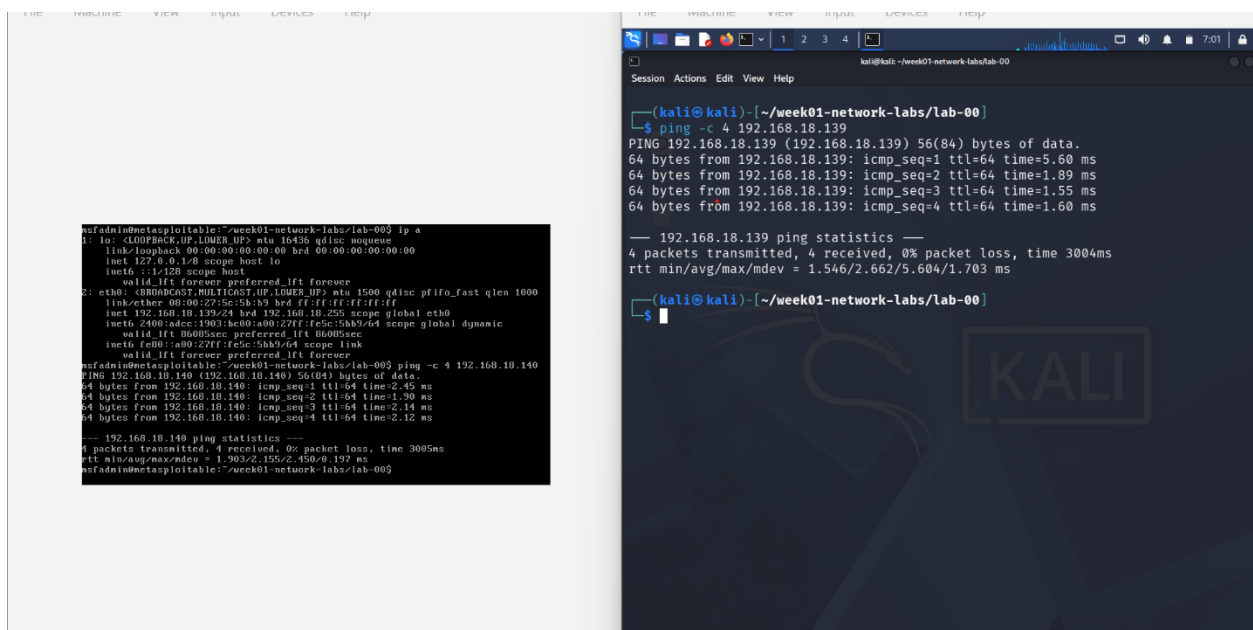
(msfadmin@metasploitable)-[~/week01-network-labs/lab-00]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        inet6 ::1/128 scope host
            valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:5c:5b:b9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.18.139/24 brd 192.168.18.255 scope global dynamic eth0
        inet6 2400:adcc:1903:bc00:a00:27ff:fe5c:5bb9/64 scope global dynamic
            valid_lft 86005sec preferred_lft 86005sec
        inet6 fe80::a00:27ff:fe5c:5bb9/64 scope link
            valid_lft forever preferred_lft forever
```

For kali linux: inet: 192.168.18.140 broadcast address: 192.168.18.255

For metasploitable2: inet: 192.168.18.139 broadcast address : 192.168.18.255

Both addresses were assigned by the physical router's DHCP server — the same DHCP server that assigns addresses to your phone, laptop, and every other real device on your WiFi. The VMs appear to the router as ordinary devices with their own MAC addresses.

VM-to-VM test:



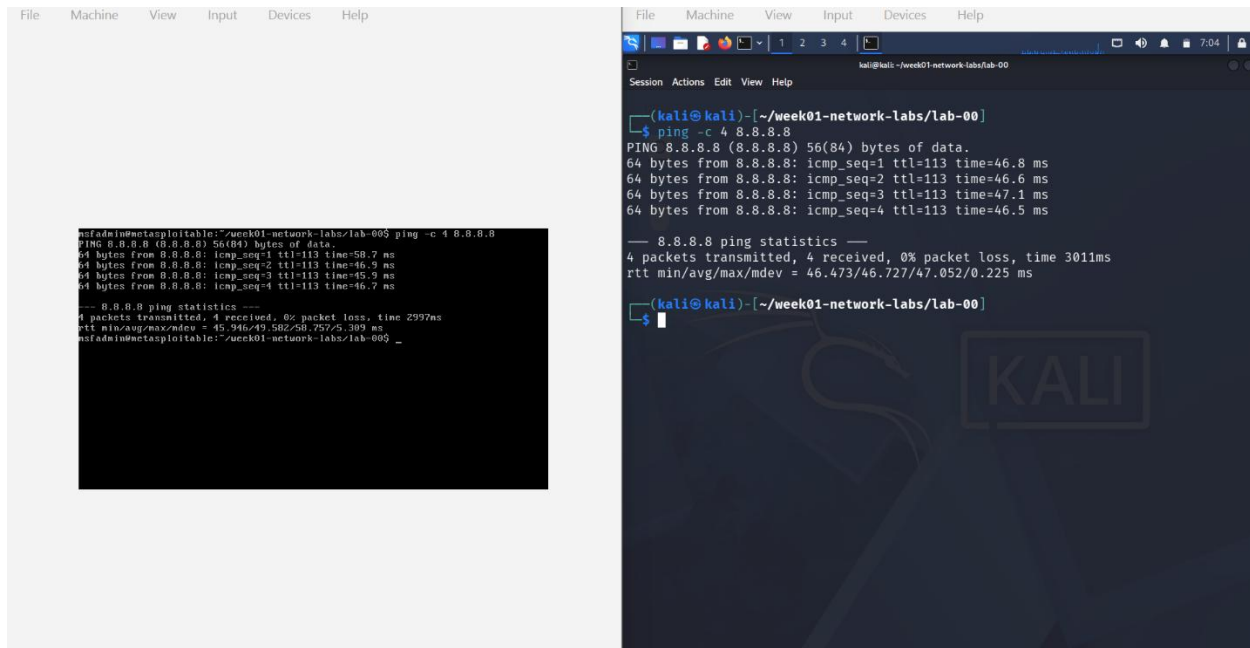
The left terminal shows the Kali Linux VM performing a ping test to the Metasploitable2 VM at IP 192.168.18.140. The right terminal shows the Metasploitable2 VM performing a ping test to the Kali Linux VM at IP 192.168.18.139. Both tests show successful results with 0% packet loss.

```
(kali@kali)-[~/week01-network-labs/lab-00]
$ ping -c 4 192.168.18.140
PING 192.168.18.140 (192.168.18.140) 56(84) bytes of data:
64 bytes from 192.168.18.140: icmp_seq=1 ttl=64 time=2.45 ms
64 bytes from 192.168.18.140: icmp_seq=2 ttl=64 time=1.90 ms
64 bytes from 192.168.18.140: icmp_seq=3 ttl=64 time=2.14 ms
64 bytes from 192.168.18.140: icmp_seq=4 ttl=64 time=2.12 ms
--- 192.168.18.140 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.903/2.155/2.450/0.197 ms

(msfadmin@metasploitable)-[~/week01-network-labs/lab-00]
$ ping -c 4 192.168.18.139
PING 192.168.18.139 (192.168.18.139) 56(84) bytes of data:
64 bytes from 192.168.18.139: icmp_seq=1 ttl=64 time=5.60 ms
64 bytes from 192.168.18.139: icmp_seq=2 ttl=64 time=1.89 ms
64 bytes from 192.168.18.139: icmp_seq=3 ttl=64 time=1.55 ms
64 bytes from 192.168.18.139: icmp_seq=4 ttl=64 time=1.60 ms
--- 192.168.18.139 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.546/2.662/5.604/1.703 ms
```

Succeeded both VMs are on the same physical LAN.

Internet test:

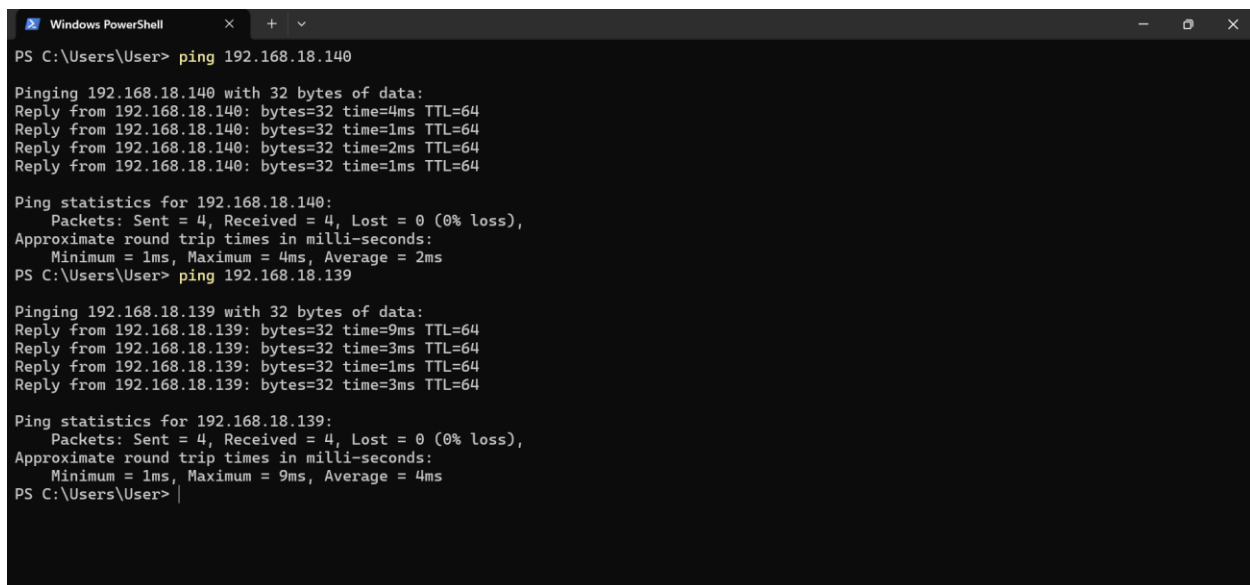


```
msfadmin@metasploitable:~/week01-network-labs/lab-00$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=113 time=58.7 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=113 time=46.9 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=113 time=45.9 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=113 time=46.7 ms
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2997ms
rtt min/avg/max/mdev = 45.946/49.582/58.757/5.389 ms
msfadmin@metasploitable:~/week01-network-labs/lab-00$ _

kali@kali:~/week01-network-labs/lab-00$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=113 time=46.8 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=113 time=46.6 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=113 time=47.1 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=113 time=46.5 ms
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3011ms
rtt min/avg/max/mdev = 46.473/46.727/47.052/0.225 ms
kali@kali:~/week01-network-labs/lab-00$
```

Succeeded with noticeably higher latency (40–100ms) compared to previous modes. This is because packets now travel through the real physical router to reach the internet, rather than through VirtualBox's internal NAT engine. The higher latency is real-world network latency.

Host to VM test:



```
PS C:\Users\User> ping 192.168.18.140

Pinging 192.168.18.140 with 32 bytes of data:
Reply from 192.168.18.140: bytes=32 time=4ms TTL=64
Reply from 192.168.18.140: bytes=32 time=1ms TTL=64
Reply from 192.168.18.140: bytes=32 time=2ms TTL=64
Reply from 192.168.18.140: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.18.140:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms
PS C:\Users\User> ping 192.168.18.139

Pinging 192.168.18.139 with 32 bytes of data:
Reply from 192.168.18.139: bytes=32 time=9ms TTL=64
Reply from 192.168.18.139: bytes=32 time=3ms TTL=64
Reply from 192.168.18.139: bytes=32 time=1ms TTL=64
Reply from 192.168.18.139: bytes=32 time=3ms TTL=64

Ping statistics for 192.168.18.139:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 9ms, Average = 4ms
PS C:\Users\User>
```

Tested, and inherently succeeds all three devices (host, Kali, Metasploitable2) are peers on the same physical LAN and can reach each other directly.

Critical security warning: Metasploitable2 is intentionally vulnerable. In Bridged mode it is directly visible to every device on your real network your phone, other computers, your router. Anyone or anything on that network could potentially interact with it. Never leave Metasploitable2 in Bridged mode unattended or on a shared/public network.

6. Overall Comparison Table

Mode	VM-to-VM	Internet	Host → VM	IP Assignment
NAT	✗ (self-loop)	✓	✗	Automatic, identical per VM
NAT Network	✓	✓	✗	Automatic, shared pool
Host-Only	✓	✗	✓	Automatic (DHCP built-in)
Internal Network	✓ (after manual config)	✗	✗	Manual (no DHCP)
Bridged	✓	✓	✓ (real LAN peer)	Automatic, real router DHCP

7. Conclusion and Decision Making Based on Labs

This investigation showed that network mode selection is a foundational design decision, not a cosmetic setting. The wrong choice can make an entire lab impossible (NAT for a multi-machine attack) or introduce real risk (Bridged with Metasploitable2 on shared WiFi).

For future labs, the selection guide is:

- **Internal Network:** all packet capture, ARP, ICMP, TCP, DNS, and scanning labs. Clean isolated traffic, no noise, no risk of exposure. This is the default for all offensive labs in this curriculum.
- **NAT Network:** any lab that needs VM-to-VM communication AND internet access simultaneously (e.g. pulling a tool on Kali while also running scans against Metasploitable2).
- **Host-Only:** any lab where you want to connect to a VM service from your Windows host (e.g. opening Metasploitable2's web interface in your Windows browser, or SSH-ing into Kali from Windows Terminal).

- **NAT:** single-VM labs only, or when a VM needs internet access with no interaction with other VMs.
- **Bridged:** use only when explicitly required and only briefly. Always shut down Metasploitable2 before switching it to Bridged.